

Day 02 Task Report

Title: Packet Sniffing, Firewall Configuration, and Vulnerability Scanning

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Environment: Kali Linux 2025 (VM)

1. Objective

The purpose of this task was to explore three key areas of network and system security:

1. **Packet Sniffing** – Capture and analyze live network traffic.
 2. **Firewall Configuration** – Secure the host system using firewall rules.
 3. **Vulnerability Scanning** – Deploy and run OpenVAS/Greenbone scanner to identify system vulnerabilities.
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2. Tools & Technologies Used

- Kali Linux 2025
 - Python3 (for custom packet sniffer)
 - iptables / UFW (for firewall configuration)
 - Docker + OpenVAS (mikesplain/openvas) (for vulnerability scanning)
 - Network utilities: tcpdump, netstat, ip a
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3. Packet Sniffing

3.1 Steps Performed

1. Created a custom packet_sniffer.py script in Python using socket library.
2. Executed the script with elevated privileges:
3. `sudo python3 packet_sniffer.py`
4. Observed live packet capture on interface eth0.
5. Captured packet headers and protocol details (IP, TCP, UDP, ICMP).

3.2 Observations

- Successfully intercepted ICMP (ping) and TCP packets.
- Verified packet details such as source IP, destination IP, and protocol type.

3.3 Screenshots

```
(kali㉿kali)-[~]
$ python3 -m scapy
INFO: Can't import PyX. Won't be able to use psdump() or pdfdump().

      aSPY//YASa
    apyyyyCY/////////YCa
  sY////////YSpcs  scpCY//Pp
ayp ayyyyyySCP//Pp      syV//C
AYAsAYYYYYYYY//Ps      cY//S
  pCCCCY//p      cSSps y//Y
  SPPPP//a      pP//AC//Y
    A//A      cyP///C
    p///Ac      sC///a
    P///YCpc      A//A
  scccccp///pSP///p      p//Y
  sY/////////y caa      S//P
  cayCyayP//Ya      pY/Ya
  sY/PsY///YCc      aC//Yp
  sc sccaCY//PCypaapyCP//YSs
    spCPY/////////YPSps
      ccaacs

| Welcome to Scapy
| Version 2.6.1
| https://github.com/secdev/scapy
| Have fun!
| Craft me if you can.
| -- IPv6 layer

>>> using IPython 8.30.0
```

```
(kali㉿kali)-[~]
$ sudo python3 packet_sniffer.py
Captured Protocols: Counter({'TCP': 84, 'UDP': 16})
```

4. Firewall Configuration

4.1 Steps Performed

1. Listed default rules:
2. `sudo iptables -L -v`
3. Configured rules to allow SSH (22) and block ICMP (ping):
4. `sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT`
5. `sudo iptables -A INPUT -p icmp -j DROP`
6. Saved firewall rules for persistence.

4.2 Observations

- Verified blocked ICMP using ping command.
- Confirmed SSH was still accessible.

4.3 Screenshots

```
(kali@kali)-[~]
$ sudo iptables -L -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination

(kali@kali)-[~]
$ sudo iptables -L -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination      tcp dpt:ssh
 0    0 ACCEPT    tcp -- any    any    anywhere                anywhere
 0    0 ACCEPT    tcp -- any    any    anywhere                anywhere      tcp dpt:http
 0    0 DROP     all -- any    any    anywhere                anywhere

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination
```

5. Vulnerability Scanning with OpenVAS

5.1 Installation Approach

1. Initially attempted installation via apt (gvm, gsa, gvmd). This was slow due to updates.
2. Switched to Docker-based deployment for faster setup:
3. `sudo apt install -y docker.io`
4. `sudo systemctl start docker`
5. `sudo systemctl enable docker`
6. `sudo docker run -d -p 9392:9392 --name openvas mikesplain/openvas`
7. Verified container status:
8. `sudo docker ps`

Output confirmed container was running and mapped to port 9392.

5.2 Access Attempt

- Checked system IPs using:
- `ip a | grep inet`
- Attempted access to Greenbone Security Assistant (GSA) via browser:
- `https://127.0.0.1:9392`

- Faced an initial "Connection was reset" error. Troubleshooting included:
 - Reviewing container logs (sudo docker logs -f openvas).
 - Checking port binding inside container.
 - Confirming the need to use HTTPS with self-signed cert.

5.3 Observations

- Container successfully launched but UI was not immediately accessible due to startup delays/self-signed certificate issues.
- Confirmed exposure of port 9392 from container to host.
- Documented troubleshooting steps for transparency.

5.4 Screenshots

```
(kali㉿kali)-[~]
$ sudo apt install -y gvm

gvm is already the newest version (25.04.0).
The following packages were automatically installed and are no longer required:
crackmapexec libhdf5-103-1t64 libutempter0
firebird3.0-common libhdf5-hl-100t64 libwebRTC-audio-processing1
firebird3.0-common-doc libicu-dev libx265-209
icu-devtools libjxl0.10 libxnnpack0
imagemagick-6.q16 libjxl0.9 linux-image-6.11.2-amd64
libabsl20230802 liblbfgsb0 openjdk-23-jre
libbfiol libldap-2.5-0 openjdk-23-jre-headless
libc++1-19 libmagiccore-6.q16-7-extra python3-aioconsole
libc++abi1-19 libmagiccore-6.q16-7t64 python3-appdirs
libcapstone4 libmagicwand-6.q16-7t64 python3-dunamai
libconfig++9v5 libmbdcrypto7t64 python3-nfsclient
libconfig9 libmsgpack-0-1 python3-packaging-whl
libdirectfb-1.7-7t64 libnetcdf19t64 python3-poetry-dynamic-versioning
libdnnl3 libopenh264-7 python3-pyinstaller-hooks-contrib
libegl-dev libpaper1 python3-pywebview
libflac12t64 libpoppler140 python3-setproctitle
libfmt9 libpoppler145 python3-tomlkit
libfuse3-3 libpython3.12-dev python3-wheel-whl
libgdal35 libpython3.12-minimal python3.12
libgeos3.13.0 libpython3.12-stdlib python3.12-dev
libgl1-mesa-dev libpython3.12t64 python3.12-minimal
libglapi-mesa libqt5ct-common1.8 python3.12-tk
libgles-dev libqt5sensors5 python3.12-venv
libgles1 libqt5webkit5 ruby-zeitwerk
libglvnd-core-dev libqt5x11extras5 ruby3.1
libglvnd-dev libsuperlu6 ruby3.1-dev
libgtksourceview-3.0-1 libtag1v5 ruby3.1-doc
libgtksourceview-3.0-common libtag1v5-vanilla strongswan
libgtksourceviewmm-3.0-0v5 libtagc0
libgumbo2 libunwind-19

Use 'sudo apt autoremove' to remove them.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
```

Unable to connect

An error occurred during a connection to 10.0.2.15:9392.

- The site could be temporarily unavailable or too busy. Try again in a few moments.
- If you are unable to load any pages, check your computer's network connection.
- If your computer or network is protected by a firewall or proxy, make sure that Firefox is permitted to access the web.

Try Again

6. Challenges Faced

- **Permission issues while running Docker commands (resolved by using sudo).**
- **Connection reset when accessing GSA UI (likely due to HTTPS certificate or initialization delay).**
- **Time constraints: Initial apt-based installation was very slow, requiring a switch to Docker method.**

7. Conclusion

- **Packet Sniffing: Successfully captured and analyzed live traffic using a custom Python script.**
- **Firewall Configuration: Effectively implemented rules to allow essential services and block unwanted traffic.**
- **Vulnerability Scanning: Deployed OpenVAS container and documented troubleshooting process, though full UI access is pending.**